

PHC Pulse — UI/UX & Screen Specification

Filled · WHAT does the user see?

Purpose. Define exactly what users see, click and experience. Screens are taken from frontend/src (React).

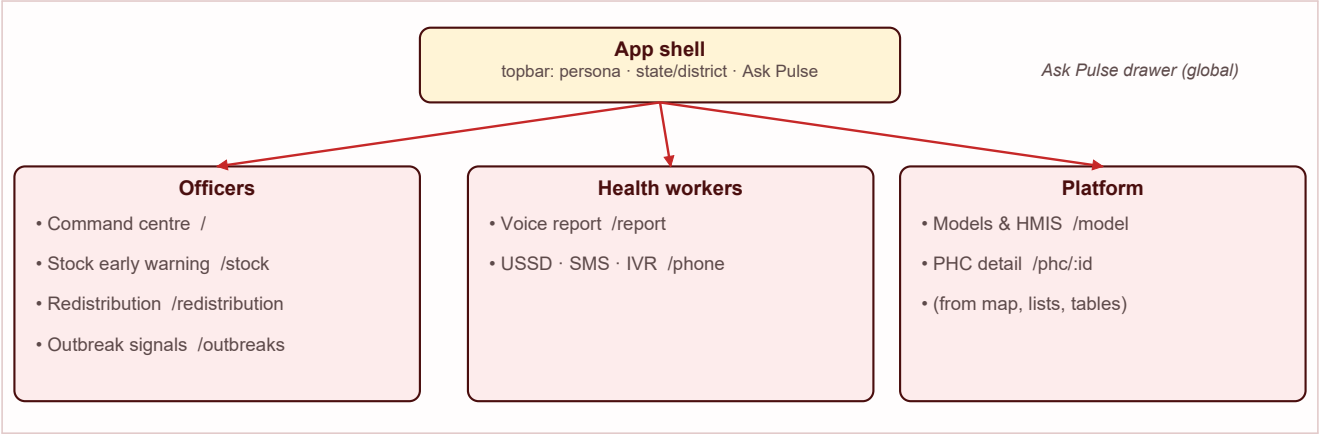
Filled from the repository PHC-Pulse-GDG-Code-for-community (main @ fd42d41) on 26 Sep 2026. Numbers quoted were measured by running the engine and the 13-test pytest suite (all passing) with PHC_TODAY=2026-09-25, seed 42.

Status legend: **[IMPLEMENTED]** in the repo and tested **[PARTIAL]** partly built **[PLANNED]** designed here, not yet in code

1. Design Principles

Principle	Definition
Action-first	Every page leads with what to do today: predicted stock-outs, transfers awaiting approval, clusters needing an IDSP alert. Recommendations carry an Approve button and a plain-language reason ("PHC-14 has surplus paracetamol; PHC-22 needs it in 9 days"). The Gemini briefing ends with 3–6 imperative actions.
Low cognitive load	Health workers get a 3-step flow (Who is reporting → Speak → Confirm). The whole report fits in 60 s and USSD needs only digits. Officers see 6 KPI tiles, one map and top-5 lists; detail is one click away. Numbers use Indian formatting (en-IN) and days, not dates.
Clear urgency	One status vocabulary everywhere: Stocked out, Critical ≤7d, High ≤14d, Watch ≤21d, OK, Surplus. Lists sort by days-to-stockout. Resilience score (0–100) as green/amber/red on the map. Clusters rank above isolated anomalies.
Human approval	The AI proposes and a person decides. Transfers go Approve → Dispatch → Delivered, each a deliberate click. Voice reports are read back and confirmed before submit. IDSP alerts are drafts. Each AI output shows an engine tag (gemini-2.5-flash or rules).
Responsive/accessibility	Grid collapses at 1100 px and 820 px; the sidebar becomes a drawer with a menu button. Min width 320 px. ARIA labels on selects and icon buttons; the drawer uses role=dialog. Tabular numerals. Noto fonts for Devanagari, Tamil, Kannada and Odia. Read-aloud (TTS) for confirmations. Planned: WCAG AA contrast audit and keyboard focus rings.

2. Information Architecture



The left sidebar is grouped by audience. A persona switcher (demo login) sets row-level scope, and the state/district filters can only narrow it. PHC detail opens from map markers, the risk matrix, warnings and transfer rows.

3. Screen Inventory

Screen	User	Purpose	Priority
Login	All roles	[PARTIAL] Demo: persona switcher in the topbar issues a signed token (national / state / DHO / PHC). Planned: Google / state SSO sign-in page mapped to a persona.	P0
Action Center	Officer	[IMPLEMENTED] Command centre: 6 KPI tiles, PHC network map (health colour, clusters), Gemini briefing (5 languages), district roll-up, weakest PHCs, top redistribution moves.	P0
PHC overview	Officer, MO	[IMPLEMENTED] /phc/:id: score, beds, staff, OPD, next supply; per-drug table; forecast chart; footfall vs expected; transfers and recent reports.	P0

Screen	User	Purpose	Priority
Inventory	Officer	[IMPLEMENTED] Stock early warning: PHC × drug risk matrix (at-risk PHCs) and early-warning list with stock, daily forecast, days to stock-out, next supply and outbreak adjustment; drug filter.	P0
Forecasts	Officer, analyst	[IMPLEMENTED] On PHC detail: stock on hand, projected stock, demand, forecast, 80% interval, today and next-supply markers. Models & HMIS: backtest KPIs and method cards.	P0
Redistribution	DHO / State / National	[IMPLEMENTED] Redistribution planner: KPIs (shipments, units, cost, escalations), lanes map, recommended transfers with Approve, cross-state toggle, escalations to district warehouse.	P0
Transfer detail	DHO	[PARTIAL] Transfer tracker on the Redistribution page: status pill, lines, km, cost, ETA/eta_at, history, Dispatch / Delivered buttons. No separate page yet.	P0
Anomaly alerts	DHO, IDSP officer	[IMPLEMENTED] Outbreak signals: cluster cards (PHCs, excess cases, max ratio, days), Draft IDSP alert (English + local), outbreak-adjusted forecasts, signal map, PHC-level anomaly table.	P1
Capacity/referral	DHO	[PARTIAL] Bed occupancy and staff attendance on the KPI tiles and PHC detail; they lower the resilience score. Referral board (free beds nearby) is planned.	P1
Voice report / Phone	ASHA, ANM, PHC staff	[IMPLEMENTED] 3-step voice report (8 languages, 60 s cap, read-back, follow-up); feature-phone simulator for USSD *123#, SMS and IVR.	P0

4. Per-Screen Specification

Screen name / purpose / data / components / actions / loading / empty / error states

Screen	Data (API)	Components & actions	Loading / empty / error
Command centre	GET /api/overview?state&district; /api/briefing?language	Kpi ×6, Leaflet MapView, Briefing (language select, Generate), district table, weakest PHCs, top moves → Redistribution	Spinner "Loading..."; "No unusual footfall today", "No transfers needed"; ErrorBox with the API message
Stock early warning	GET /api/stock?status	Risk matrix (StatusPill cells), warnings table, drug filter; row → PHC detail	"No PHCs at risk in this scope", "No early warnings"
Redistribution	GET /api/redistribution?cross_state; POST .../{rec}/approve; POST /api/transfers/{id}/status	KPIs, lanes map, recommendations with Approve, tracker with Dispatch / Delivered, escalations; toast on success	"Solving the redistribution MILP..."; "No transfers needed..."; "No transfers yet. Approve a recommendation above"; 409 stale → refresh prompt; 403 shown
Outbreak signals	GET /api/alerts; POST /api/clusters/{id}/alert	Cluster cards, Draft IDSP alert (EngineTag), outbreak-adjusted forecast list, signal map, anomaly table	"No clusters detected in this scope", "No anomalies"
PHC detail	GET /api/phcs/{id}	Kpis, drug table, forecast ComposedChart, footfall vs threshold chart, transfers and reports	404 "PHC not found in your scope"; "No voice/USSD/SMS reports yet today"
Voice report	POST /api/reports/voice /parse; POST /api/reports/submit	PHC and language pickers, prompt playback, record (60 s cap, timer), transcript, translation, editable field chips, confidence, read-aloud, follow-up answer, Submit	Recording/processing spinners; "No speech captured..." hint; 503 Gemini message with text-mode fallback
USSD · SMS · IVR	POST /api/ussd, /api/sms, /api/dialogflow/webhook	Feature-phone keypad simulator, SMS composer, IVR form in a chosen language	Unregistered number → END message; bad SMS → format hint

5. Design System

Item	Decision
Brand/colors	Maroon sidebar #4a0e0e (700 #5c1515), primary red #c62828, orange accent #ef6c00, blush background #fbf5f4, cards #ffffff with 14 px radius and a soft maroon shadow, ink #2b1414 / #5d4a4a / #8c7a7a, lines #efdcde. Brand mark: pulse icon on a red rounded square.
Typography	Inter / Segoe UI with Noto Sans (+ Devanagari, Tamil, Kannada, Oriya) fallbacks. Page H1 24 px, card titles 15 px, KPI values 28 px / 750 weight, body 13–14 px, eyebrow 11 px uppercase red. Tabular numerals for figures.

Item	Decision
Status colors	Stocked out #fde0e0/#a01818 · Critical #ffe1d1/#b33c00 · High #fff4d6/#8a5600 · Watch #fff7cc/#7a6500 · OK / Delivered #e6f4e7/#2e7d32 · Surplus / In transit #e3eeff/#1565c0. Health: green #2e7d32, amber #e69100, red #c62828.
Charts	Recharts ComposedChart: stock (blue area), projected stock (orange dashed area), 80% interval (pink band), demand (maroon line), forecast (red line), reference lines for today and next supply; footfall vs dashed orange alert threshold. Leaflet maps with health-coloured markers, cluster rings and transfer lanes. Animations off.
Buttons/forms	.btn (white, 1 px line, 600 weight), .btn.primary (red), .btn.ghost, .btn.sm; .select/.input with 9 px radius; pills for status and engine tags; toast for confirmations; right drawer for Ask Pulse. Destructive or irreversible steps are separate explicit buttons.